

Introduction to Computer Vision

Lecture 04 - Linearity and Convolution

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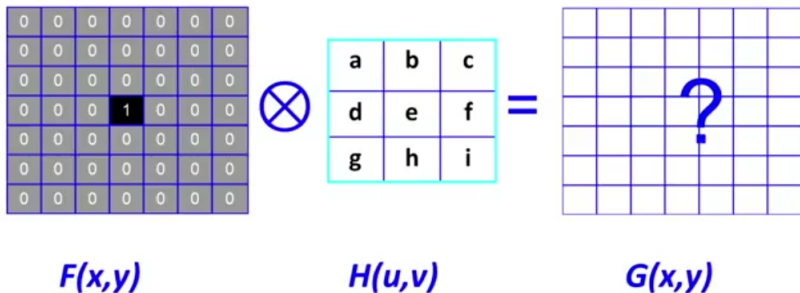
Contents

- 1 Filtering an image
- 2 Correlation vs Convolution
 - Correlation
 - Convolution
- 3 Boundary problems
- 4 Linear filters

Convolution

Filtering an image

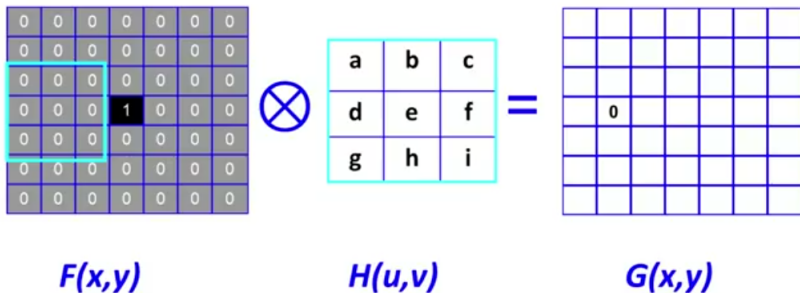
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Convolution

Filtering an image

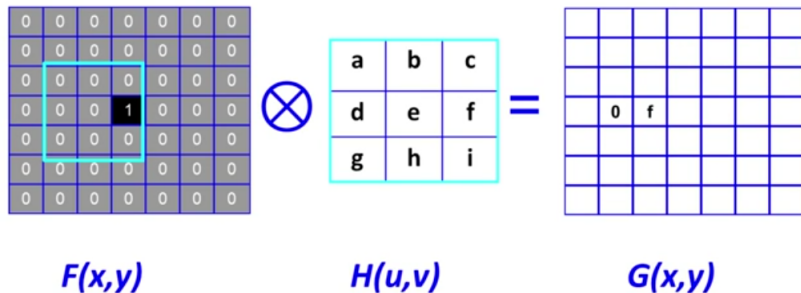
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Convolution

Filtering an image

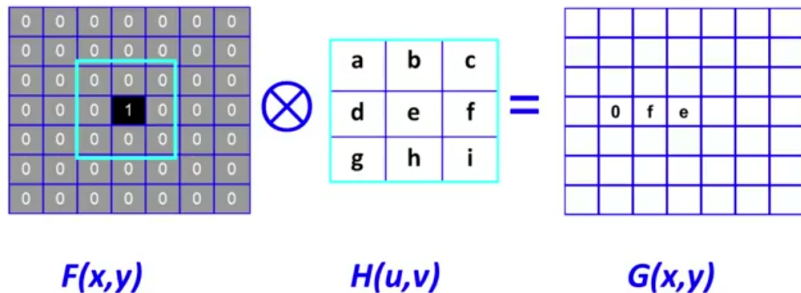
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Convolution

Filtering an image

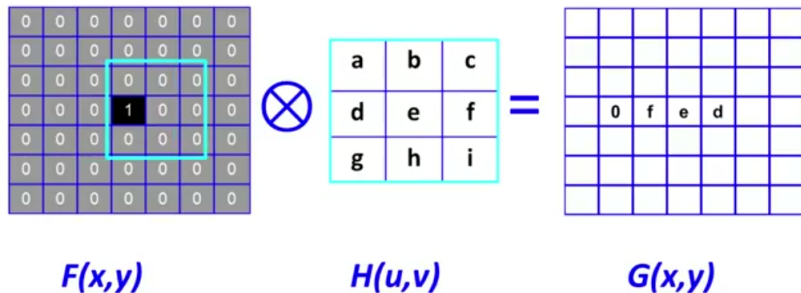
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Convolution

Filtering an image

•



Convolution

Filtering an image

•

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

$F(x,y)$



a	b	c
d	e	f
g	h	i

$H(u,v)$

=

	0	i	h	g		
	0	f	e	d		
	0	c	b	a		

$G(x,y)$

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Correlation vs Convolution

Correlation

$$f \otimes h = \sum_k \sum_l f(k, l) h(i + k, j + l)$$

f = Image

h = Kernel

f

f_1	f_2	f_3
f_4	f_5	f_6
f_7	f_8	f_9

$\otimes$

h

h_1	h_2	h_3
h_4	h_5	h_6
h_7	h_8	h_9

$\rightarrow$

$$\begin{aligned} f * h &= f_1 h_1 + f_2 h_2 + f_3 h_3 \\ &+ f_4 h_4 + f_5 h_5 + f_6 h_6 \\ &+ f_7 h_7 + f_8 h_8 + f_9 h_9 \end{aligned}$$

Correlation vs Convolution

Correlation

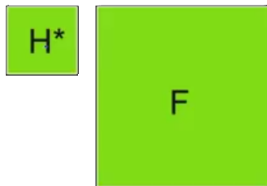
- 1D correlation.

$$H \otimes F(i) = \sum_{k=-N}^N H(k)F(i+k)$$

- 2D correlation.

$$G(i,j) = \sum_{k=-n}^n \sum_{l=-n}^n H[k,l]F(i+k,j+l)$$

$$G = F \otimes H$$



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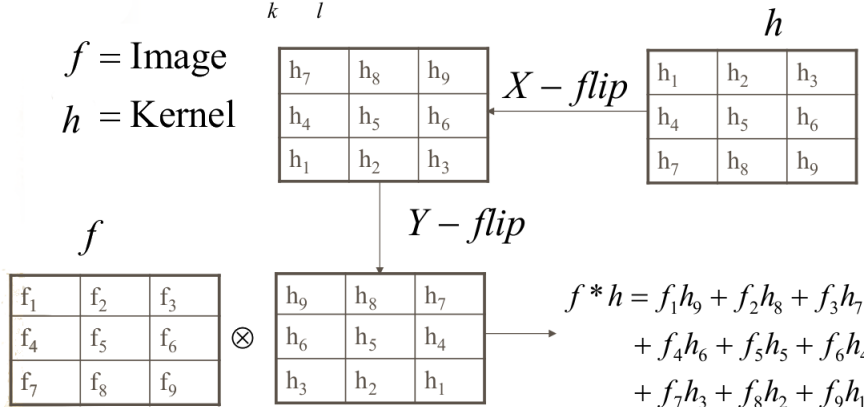
Correlation vs Convolution

Convolution

$$f * h = \sum_k \sum_l f(k, l) h(i - k, j - l)$$

f = Image

h = Kernel



Correlation vs Convolution

Convolution

- 1D convolution.

$$H * F(i) = \sum_{k=-N}^N H(k)F(i - k)$$

- 2D convolution.

$$G(i, j) = \sum_{k=-n}^n \sum_{l=-n}^n H[k, l]F(i - k, j - l)$$

$$G = F * H$$



Figure: Notation for a Convolution. Source: K. Grauman

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Boundary problems

Size of the output?

- What is the size of the output image? The shape is
 - **full**: Output size is of sizes of f and g
 - **same**: Output size is same of f
 - **valid**: Output size is **difference** of sizes of f and g

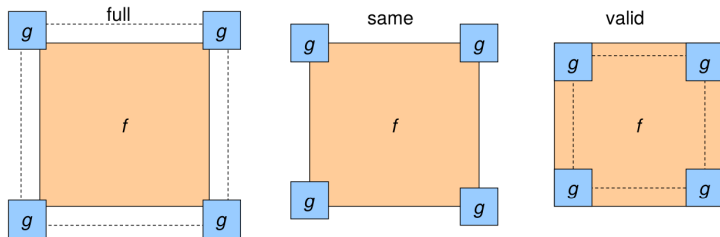


Figure: Boundary problems. Source: S. Lazebnik

Boundary problems

Size of the output?

- clip filter (black)



Boundary problems

Size of the output?

- clip filter (black)



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Linear Filters



Original

0	0	0
0	0	1
0	0	0

?

Linear Filters

- Filter with no change.



Original

0	0	0
0	0	1
0	0	0



Shifted left
By 1 pixel

Linear Filters



Original

0	0	0
0	2	0
0	0	0

-

$\frac{1}{9}$

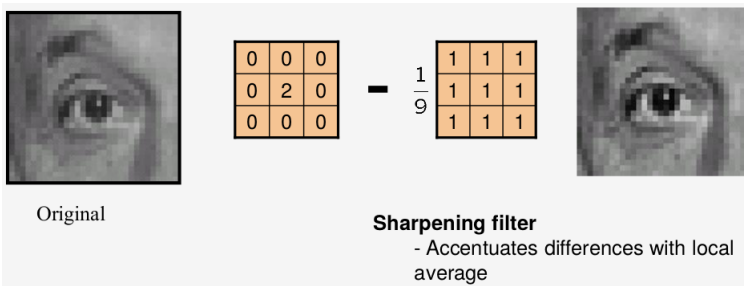
1	1	1
1	1	1
1	1	1

?

(Note that filter sums to 1)

Linear Filters

- Filter with no change.



Original

0	0	0
0	2	0
0	0	0

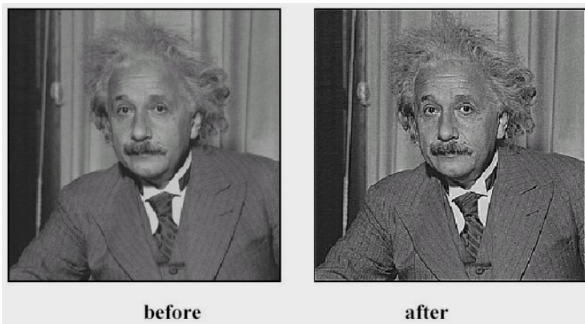
— $\frac{1}{9}$

1	1	1
1	1	1
1	1	1

Sharpening filter
- Accentuates differences with local average

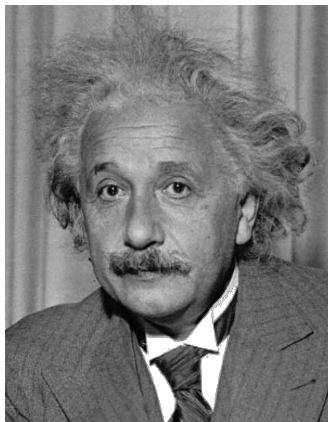
Linear Filters

- Filter with no change.



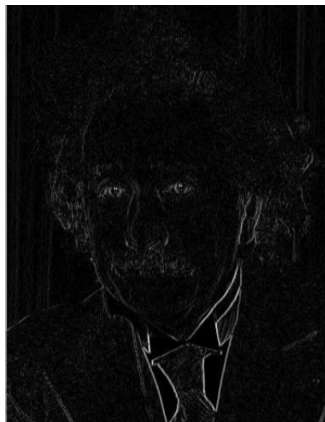
Linear Filters

- Filter with no change.



1	0	-1
2	0	-2
1	0	-1

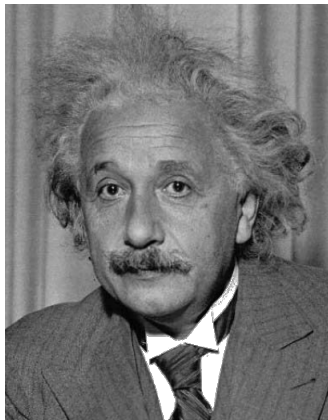
Sobel



Horizontal Edge
(absolute value)

Linear Filters

- Filter with no change.



1	2	1
0	0	0
-1	-2	-1

Sobel



Horizontal Edge
(absolute value)

The End